$$\frac{\Gamma \vdash e : \tau \rightarrowtail e_1}{\Gamma \vdash \mathsf{new}(e) : \mathsf{ref}(\tau) \rightarrowtail \mathsf{newcell}\ e_1}\ \text{Compile-New}$$

$$\frac{\Gamma \vdash e : \mathsf{ref}(\tau) \rightarrowtail e_1}{\Gamma \vdash \mathsf{get}(e) : \tau \rightarrowtail \mathsf{getcell}\ e_1}\ \text{Compile-Get}$$

$$\frac{\Gamma \vdash e_1 : \mathsf{ref}(\tau) \rightarrowtail e_3 \qquad \Gamma \vdash e_2 : \tau \rightarrowtail e_4}{\Gamma \vdash \mathsf{set}(e_1, e_2) : \tau \rightarrowtail \mathsf{setcell}\ e_3\ e_4}\ \text{Compile-Get}$$

$$\frac{\begin{array}{c} x \notin \Gamma \\ \Gamma, x : \langle \mathsf{unit} \to \tau, \tau \to \mathsf{loc} \rangle \vdash \mathsf{mon}_j^{k,l}(\langle \mathsf{any} \to \kappa, \kappa \to \mathsf{any} \rangle, x) : \langle \mathsf{unit} \to \tau, \tau \to \mathsf{loc} \rangle \rightarrowtail e_1 \\ \Gamma \vdash e : \mathsf{ref}(\tau) \rightarrowtail e_2 \qquad e_3 = e_1[x := e_2] \end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{ref/c}(\kappa), e) : \mathsf{ref}(\tau) \rightarrowtail e_3}\ \text{Compile-Mon-Ref}$$

$$\begin{array}{lcl} \texttt{newcell} & = & \lambda v.(\lambda l.\langle \lambda_.\ \mathsf{get}(l), \lambda v.\ \mathsf{set}(l, v) \rangle)\ \mathsf{new}(v) \\ \texttt{getcell} & = & \lambda b.\ \mathsf{inj.l}(b)\ \mathsf{unit} \\ \texttt{setcell} & = & \lambda b.\lambda v.(\lambda_.b\ (\mathsf{inj.r}(b)\ v)) \end{array}$$

Listing 1: Compiler Rules for Mutable Cells